

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI
For

MASTERS IN DISASTER MANAGEMENT

(Two Year Degree Programme, 1st & 2nd Semester System)

NEP 2020

Departments of Geography, Panjab University Chandigarh

(Implemented from the Academic Session 2026-27)

Examinations, 2026-27

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APPLICABILITY OF REGULATIONS FORTHE TIME BEING IN FORCE

Notwithstanding the integrated nature of a programme spread over more than one academic year, the regulations in force at the time a student joins the course shall hold good only for the examinations held during or at the end of the academic year. Nothing in these regulations shall be deemed to debar the University from amending the regulations subsequently and the amended regulations, if any, shall apply to all students whether old or new.

ABOUT THE COURSE

Masters in Disaster Management is a professional two years postgraduate programme offered to innovative and motivated minds from across the academic and professional fields who are inclined to contribute in creating disaster resilientand safe society. The Department of Geography, Panjab University Chandigarhhas been the pioneer in introducing professional programmes ofcontemporary relevance. We first introduced a post-graduate degree programme in Disaster Management in the year 2008 which since then has been restructured, upgraded and updated with changing disaster scenario, management paradigms, strategies and policy frameworks.Keeping pace with changing global, national and local disaster scenario and recognising the need to develop our society in a sustainable manner, thisprogramme offers innovative, multi-disciplinaryand robust course content that caters to the needs of different aspects and dimensions of disaster management and its outreach. This two-yearprogramme divided into four semesters is designed to inculcate professional attitude with sound understanding of theoretical, conceptual and field-based knowledge so that the students can fulfil the increasing demand for skilled professionals to deliver proficiently in disaster risk reduction and management.

WHY STUDY WITH US

Department of Geography, Panjab University offers uniquely designed dynamic professional programme that not only produces highly trained professionals but also instilsa value system in its students so that they contribute in making this nation resilient. During the two years stay with us, students are exposed to a wide range of knowledge-learning and character-building practices. We offer a wide spectrum of learning skills ranging from conceptual and theoretical clarity of the subject, technical capabilities, analytical capacities, understanding of social, economic, political and humanitarian contexts of disaster mitigation, preparedness, response, risk reduction and disaster proofing. Trained through a focussed field-work, learning by doing approach and research work, we prepare our students to develop practical understanding of processes, mechanisms and operational aspects of disaster management.

Some of the highlights of our curriculum are:

- Conceptual and theoretical underpinnings of disaster risk and climate change studies.
- Understanding of mechanisms and processes of hazards, disasters and climate change.
- Critical evaluation of global and regional disaster manifestation and climate change.
- Understanding of disaster risk reduction procedures, practices and policies.
- Sensitization on social and cultural implications of disasters and climate change.
- Training on hazard zonation, risk analysis, and disaster response.
- Inculcating disaster risk reduction practices through community participation.
- Technical skill development through geospatial technology.
- Building partnership with the government and other stakeholders
- Develop research aptitude for project management.
- Formulate public policy and planning mechanisms for risk reduction

ELIGIBILITY CRITERIA

- The eligibility for admission to the course shall be as follows:
  - Graduate degree in any discipline with at least 50% marks in aggregate from any University recognized by the Panjab University as equivalent thereto. The admission to the course shall be through an Entrance Test.
  - All candidates, except those sponsored by the Government/Semi-Government Organisations and Institutions, will be required to qualify the Entrance Test conducted by the Panjab University, Chandigarh. The apportioning of weightages for the purpose of preparing the merit list will be as follows:

    Entrance Test: 50%

    Qualifying Examination (Bachelor's Degree): 50%

    Academic and other weightage(s), if any, shall be based on the percentage of marks obtained by the eligible candidates in the Qualifying Examination (Bachelor's Degree) as prescribed and admissible under Panjab University Rules.

GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%)

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of students:
   - Terminal Evaluation:80% (or as specified in the syllabus)
   - Continuous Assessment: 20 %(or as specified in the syllabus)
   - Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.
   - In order to incorporate an element of Continuous Internal Assessment of students, the Colleges/Departments will conduct the following:
     - Written Test: 25
     - Snap Test: 25
     - Participation in Class discussion: 15
     - Term Paper: 25
     - Attendance: 10

     Total: 100 (reduced to 20/10 or as specified in the syllabus)
2. Weightage of 2/1 mark(s) for attendance component out of 20/10 marks for continuous assessment shall be available only to those students who attend 75% and more of classroom lectures/seminars/workshops. The break-up of marks for **attendance component** for theory papers shall be as under:

Attendance Component	***Mark/s for Theory Papers***
(a) 75 % and above up to 85 %:	1
(b) Above 85 %:	2

3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20/10, will be carried forward and added to his/her score out of the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examinations.
4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the *proforma* obtainable from the Examination Branch.

PROGRAMME OUTLINE

SEMESTER-I

CourseType	Course Code	Course Title	Max. Marks	Credit	Hours per Week
DSC (Theory)	DM 101	Foundations of Disaster Management	100	4	4
	DM 102	Introduction to Hazards and Disasters	100	4	4
	DM 103	Climate Change and Development Interface	50	2	2
	DM 104	Hazards and Disasters in India	50	2	2
DSC (Practical)	DM 105	Hazard, Vulnerability and Risk Assessment	50	2	4
	DM 106	Mapping Disasters in India	50	2	4
	DM 107	Disaster Emergency Management Skills	50	2	4
	DM 108	Statistical Analysis	50	2	4
		Semester Total	***500***	***20***	

SEMESTER-II

CourseType	Course Code	Title	Max. Marks	Credit	Hours per Week
DSC (Theory)	DM 201	Disaster Management Mechanisms in India	100	4	4
	DM 202	Global Initiatives for DRR and Climate Resilience	100	4	4
	DM 203	Fundamentals of Geoinformatics	50	2	2
RBC-RM	DM 204	Research Methodology in Disaster Studies	50	2	2
	DM205	Research Tools and Techniques	50	2	4
DSC (Practical)	DM 206	Disaster Preparedness and Capacity Building	50	2	4
	DM 207	Geospatial Tools and Techniques	50	2	4
	DM 208	Field-work Based Activity	50	2	4
		Semester Total	***500***	***20***	

SEMESTER-III

CourseType	Course Code	Title	Max. Marks	Credit	Hours per Week
DSC (Theory)	DM 301	Urban Disasters and Risk Reduction	50	2	2
	DM 302	Incident Response System	50	2	2
	DM 303	Community Based Disaster Risk Management	50	2	2
	DM 304	Geoinformatics for Disaster Studies	50	2	2
RBC-RM	DM 305	Qualitative Research Methods	100	4	2
DSC (Practical)	DM 306	Disaster Awareness and Knowledge Dissemination	50	2	4
	DM 307	Geoinformatics for Disaster Management	50	2	4
	DM 308	Internship and Internship Report	100	4	2+4*
		Semester Total	***500***	***20***	

SEMESTER-IV

CourseType	Course Code	Title	Max. Marks	Credits	Hours per Week
RBC-DST	DM 401	Dissertation 1. Formulation of Research Proposal 2. Mid-Term appraisal 3. Pre-submission Presentation 4. Dissertation Evaluation 5. Final Presentation 6. Viva-voce	 50 75 75 200 50 50	20	2+4*
		Semester Total	***500***	***20***	
		GRAND TOTAL (Semester-I, II, III, IV)	**2000**	**80**	

*4 hours per teacher supervising the dissertation/internship report

NOTE:

1. The minimum marks required to pass the examination shall be 50% in aggregate in each semester and 45% in each individual paper: (i) theory paper, (ii) practical examination, (iii) internship report, (iv) Research Report, and (v) dissertation.
2. A candidate, who fails in a particular semester, shall be allowed to re-appear in the subsequent semester with a maximum of two chances to be availed consecutively for the respective semester.
3. A candidate who fails in one or more theory and/or practical/internship report/research report/dissertation examination/s shall be allowed to re-appear in the subsequent semester with a maximum of two chances to be availed consecutively in the respective theory and/or practical/internship report/research report/dissertation examination/s.
4. The Internship Report shall be prepared by the candidate on the basis of work done or experience gained on visit(s) to Disaster Management Centre(s)/Institute(s)/NGO(s) in India. The candidate shall submit three copies of his/her Internship Report ten days before the commencement of the theory examination of the said semester.
5. In case of Research Report/Dissertation, the Board of Control shall assign Supervisor/Co-supervisors for each candidate for supervising candidate's dissertation on an approved topic. External Supervisor/Co-supervisors may be allotted, subject to their eligibility to be determined by the Board of Control.

SEMESTER-I

DM 101
FOUNDATIONS OF DISASTER MANAGEMENT

Course Code: DM 101	**Max. Marks: 100**
Course Type: DSC- Theory	Theory Paper: 80
Course Credit: 4	Internal Assessment: 20
Teaching Hours/Week: 4	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Define and relate to key concepts, philosophy and scope of disaster management.
- Explain and summarise models and approaches to disaster management.
- Apply, compare, and deduce elements, phases, and activities of disaster management cycle.
- Generalise, synthesize and critique disaster response and crisis management activities.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each part shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to the compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

Basic Concepts of Disaster Management

i. Disaster Management:Definition, Scope and Significance
ii. Elements of Disaster Management; Disaster Management Cycle
iii. Disaster Management Models w.r.t. PAR and Access Model

Unit-II

Disaster Mitigation, Prevention and Preparedness

iv. ***Mitigation:*** Knowledge of disaster specific risk; analysing disaster damages, interventions for minimizing disaster impacts; preparation of mitigation plan.
v. ***Prevention:*** Analysing the nature of hazards, ways of minimizing hazard intensity; preparation of disaster prevention plan.
vi. ***Preparedness:*** Awareness generation; information management; early warning system; community participation; training and capacity building; preparedness plan preparation.

Unit-III

Disaster Emergency Response and Crisis Management

vii. ***Crisis Management:***Rescue, relief, rehabilitation & reconstruction; crisis management plan.
viii. ***Emergency Response:*** Standard operation procedure (SOP); information management system; warning dissemination; evacuation; search and rescue operations.
ix. ***Relief Operations:***Arranging for shelter, food, safe drinking water, sanitation and medical aids; maintaining law and order; emergency operation center (EOC)

Unit-IV

Recovery, Rehabilitation and Reconstruction

x. ***Disaster Recovery:***Actions for rehabilitation and reconstruction; encouraging and facilitating necessary adjustments to reduce disaster risk.

xi. ***Disaster Rehabilitation:*** Damage assessment; need analysis; resource mobilization; restoration of basic amenities; creating livelihood options.

xii. ***Disaster Reconstruction:***Repair and reconstruction of infrastructure; restoration of operations of the service sector.

LIST OF READINGS

Essential Readings:

1. Andrew E. C., Bernard M., Janaka J., Jones S. (eds.). *Hazards, Risks, and Disasters in Society*. Netherlands, Elsevier Science, 2014.
2. Bird, R. (ed.). *Handbook of Natural Hazards and Disasters*. United States, Larsen and Keller Education, 2022.
3. Coppola, Damon P.: *Introduction to International Disaster Management*. Elsevier, London, 2015.
4. Bankoff, G., Hilhorst, D. and Frerks, G. *Mapping Vulnerability: Disasters, Development and People*. United Kingdom: Taylor & Francis Group. 2013.
5. Madu, C. N. *Handbook of Disaster Risk Reduction & Management: Climate Change and Natural Disasters*. Singapore: World Scientific Publishing Company. 2017.

Suggested Readings:

1. Anderson M. *Disaster Vulnerability and Sustainable Development: A General Framework for Assessing Vulnerability*, Cambridge, 1993.
2. Anderson, M. and P. Woodrow. *Rising from the Ashes: Development Strategies in Times of Disaster*. UNESCO and West view Press, Inc., Colorado, 1989.
3. Burton, I., Kates, R.W. and White, G.F. *Environment as Hazard*, Guilford Press, New York, 1993.
4. Chakraborty, S.C. *Natural Hazards and Disaster Management*, Pragatishil Prakashak, Kolkata, 2007.
5. David Godschalk, Timothy Beatley, Philip Berke, David Brower, Edward J. Kaiser. *Natural Hazard Mitigation: Recasting Disaster Policy and Planning*, Island Press, 1998.

DM 102
INTRODUCTION TO HAZARDS AND DISASTERS

Course Code: DM 102	**Max. Marks: 100**
Course Type: DSC- Theory	Theory Paper: 80
Course Credit: 4	Internal Assessment: 20
Teaching Hours/Week: 4	Examination Time: 3 Hours

COURSE OUTCOMES: The course would enable students to:

- Acquire conceptual clarity of hazard, vulnerability, risk and disaster concepts.
- Recognise and classify types of hazards and disasters.
- Examine and measure vulnerability and disaster risk.
- Synthesise and apply knowledge to meaningfully interpret disaster risk.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each part shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to the compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

i. Hazard and Disaster: Concept and Characteristics
ii. Types and Classification of Hazards and Disaster
iii. Disaster Studies: Evolving Management Approaches

Unit-II

iv. Geotectonic and Geomorphological Hazards: Earthquake andLandslide
v. Hydrometeorological Hazards: Floods and Cyclones
vi. Multifaceted Hazard: Drought

Unit-III

vii. Vulnerability: Types and Elements(Exposure, Resistance, Resilience)
viii. Vulnerability Assessment Framework and Methods
ix. Disaster Risk: Concept, Characteristics, Drivers and Assessment Techniques

Unit-IV

x. Human-Made Hazards and Disasters: Types and Characteristics
xi. Terrorism as Disaster: Types and its Socio-Political-Economic dimensions
xii. Wars and Conflicts as Humanitarian Crisis: Causes, Trends and Challenges

LIST OF READINGS

Essential Readings:

1. Andrew E. C., Bernard M., Janaka J., Jones S. (eds.). *Hazards, Risks, and Disasters in Society*. Netherlands, Elsevier Science, 2014.
2. Bird, R. (ed.). *Handbook of Natural Hazards and Disasters*. US, Larsen and Keller Edu. 2022.
3. Hewitt, K. *Regions of Risk: A Geographical Introduction to Disasters.* United Kingdom, Taylor & Francis, 2014.
4. Hyndman, D. and D. Hyndman. *Natural Hazards and Disasters*. 5th edition. Cengage Learning, Boston, USA, 2016.
5. Wisner B, Blaikie P., Cannon T. and I. Davis. *At Risk: Natural Hazards, People's Vulnerability, and Disasters,* 2nd Edition, Routledge, London, 2014.

Suggested Readings

1. Alexander, D. E. *Confronting Catastrophe: New Perspectives on Natural Disasters*. Harpenden, U.K: Terra Publishing, 2000.
2. Keller, Edward A and Duane E DeVecchio.*Natural Hazards: Earth's Processes as Hazards, Disasters, and Catastrophes.* 4th Edition, Routledge, 2016.
3. Montz, Burrell E.; Graham A. Tobin and Ronald R. Hagelman. *Natural Hazards: Explanation and Integration*. Second Edition, The Guilford Press, New York, 2017.
4. Madu, Christian N and Chu-hua Kuei. *Handbook of Disaster Risk Reduction and Management*. World Scientific Publishing Co. Pvt. Ltd, Singapore, 2018.

DM 103
CLIMATE CHANGE AND DEVELOPMENT INTERFACE

Course Code: DM 103	**Max. Marks: 50**
Course Type: DSC- Theory	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the semester, the students will be able to:

- Explain conceptual and theoretical aspects of climate change and development linkages.
- Identify the drivers of climate change and analyse, compare and debate its patterns.
- Examine and predict impacts of climate change on development.
- Critically evaluate initiatives and policy framework for climate resilience.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to the compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

Foundations to Climate Change

i. Climate Change: Concept, Facts and Global Perspective
ii. Causes of Climate Change; Information Sources of Past Climatic Change

Unit-II

Climate Change and Impacts: Scientific Basis

iii. Climate Change During the Holocene;Recent and Projected Climate Trends
iv. Climate Change Impacts on Biodiversity, Water Resources, Disaster Risk

Unit-III

Climate Change and Development Interface

v. Climate Change, Development and Sustainability: Links, Concerns and Challenges
vi. Building Climate Resilience: Mitigation and Adaptation

Unit-IV

Climate Change, Development and Policy

vii. Politics of Climate Change and Development;Global North v/s South Debate
viii. India's Climate Change Policy and Initiatives

LIST OF READINGS

Essential Readings:

1. Andrew E. C., Bernard M., Janaka J., Jones S. (eds.). *Hazards, Risks, and Disasters in Society*. Netherlands, Elsevier Science, 2014.
2. Flannery, T. (2005). *The Weather Makers: The History and Future Impact of Climate Change.* Allen Lane. London.
3. Mannion, A.M. (1999). *Natural Environmental Change.* Routledge, London.
4. Newton, David E. *The Climate Change Debate: A Reference Handbook*. US, ABC-CLIO, 2020.
5. Parkinson, Claire L. (2010). *Coming Climate Crisis? Consider the past, beware the big flux.* Rowman & Littlefield Publishers, INC. Lanham, Maryland.
6. Sturman, Andrew, and Quenol, Herve. *Climate Change: Impacts and Adaptation at Regional and Local Scales.* United Kingdom, Oxford University Press, 2024.
7. Vogler, J. *Climate Change in World Politics*. United Kingdom, Palgrave Macmillan UK, 2016.

Suggested Readings:

1. Kevin Anderson and Alice Bows (2011). Beyond 'dangerous' climate change: emission scenarios for a new world. *Phil. Trans. R. Soc. A,* 369, 20–44.
2. Mann, M. E., Bradley, R. S. & Hughes, M. K. (1998). Global-scale temperature patterns and climate forcing over the past six centuries. *Nature* 392, 779-787.
3. Ramankutty, N. & Foley, J.A. (1999). Estimating historical changes in global land cover: croplands from 1700 to 1992. *Global Biogeochemical Cycles*, 13, 997–1027.
4. Richard W. Battarbee and Heather A. Binney (2008). *Natural Climate Variability and Global Warming: A Holocene Perspective.* John Wiley & Sons.
5. Romm, J. *Climate Change: What Everyone Needs to Know*. US, Oxford University Press, 2022.

DM 104
HAZARDS AND DISASTERS IN INDIA

Course Code: DM 104	**Max. Marks: 50**
Course Type: DSC- Theory	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the semester, the students will be able to:

- Recognize and classify hazards and disasters impacting India.
- Develop, compare and formulate disaster regionalisation.
- Explain drivers and determinants of disaster risk and vulnerability in India

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to the compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

i. Geography of Hazards and Disasters in India
ii. Disaster Vulnerability in India, its Drivers and Linkages

Unit-II

iii. Earthquake Prone Regions of India
iv. Landslide Hazard in India; Drought Hazard and Risk in India

Unit-III

v. Floods in India; Cloudburst and GLOF in the Indian Himalayas
vi. Cyclone and associated Hazards in India

Unit-IV

vii. Naxalism as Disaster in India: Patterns, Problems and Challenges
viii. Terrorism in India: Patterns, Drivers and Solutions

LIST OF READINGS

Essential Readings:

1. Government of India: *National Policy on Disaster Management*. National Disaster Management Authority, Ministry of Home Affairs, New Delhi, 2009.
2. Gupta, M.C.; L.C. Gupta, B.K. Tamini, Vinod K. Sharma: *Manual on Natural Disaster Management in India,* National *Disaster Management Centre, New Delhi, 2000.*
3. Pandey, R.K. *Disaster Management in India: Policies, Institutions, Practices*. India, Taylor & Francis. 2020.
4. Valdiya, K.S. *Coping with Natural Hazards: Indian Context*. India, Orient Longman, 2004.
5. Dagur, O.S. *Disaster Management: An Appraisal of Institutional Mechanisms in India*. India, Centre for Land Warfare Studies, 2011.

Suggested Readings

1. Alexander, D. E.: *Natural Disasters*. London: University College London Press and; Dordrecht and Boston: Kluwer Academic Publishers, 1993.

2. Alexander, D. E.: *Confronting Catastrophe: New Perspectives on Natural Disasters*. Harpenden, U.K: Terra Publishing, 2000.
3. Burton, I., Kates, R.W. and White, G.F.: *Environment as Hazard*, 2nd edition, Guilford Press, New York, 1993.
4. Hewitt, K. *Regions of Risk: A Geographical Introduction to Disasters*. United Kingdom, Taylor & Francis, 2014.
5. Subramanian, R.: *Disaster Management*, Vikas Publishing House Pvt. Ltd., Noida, India, 2018.

DM 105
HAZARD, VULNERABILITY AND RISK ASSESSMENT

Course Code: DM 105	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical/Lab Work	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Recognize and classify global disasters occurrences and impacts.
- Collect and interpret data on hazardous process leading to disasters.
- Measure and compute different types of vulnerabilities.
- Identify and categorise elements/determinants of disaster risk.
- Determine and critically evaluate disaster risk.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Global Disaster Profiling: Occurrence, Casualties and Damages
2. Inter-Country Disaster Manifestations and Impacts in Asia
3. Hazard Auditing: Anticipating and Identifying the threat
4. Measuring Vulnerability
    - Assessment of Physical Vulnerability
    - Analysing Social Vulnerability
    - Evaluation of Economic Vulnerability
    - Determining Overall Vulnerability
5. Assessment of disaster risk
6. Case Study: Report based on hazard, vulnerability, and risk assessment

LIST OF READINGS

Essential Readings:

1. Elliott, A. and E.V. Hsu (eds.). *The Consequences of Global Disasters*, Routledge, London, 2016.
2. Greg Bankoff, Dorothea Hilhorst, Georg Frerks (eds.). *Mapping Vulnerability: Disasters, Development and People*. United Kingdom, Taylor & Francis, 2013.
3. IFRC. https://www.ifrc.org/documents/all (Online access to various World Disaster Reports)
4. Minay, R. *Map Your Planet: Natural Disasters*. United Kingdom, Hachette Children's Group, 2022.

5. Peijun Shi, Roger E. Kasperson (eds.). *World Atlas of Natural Disaster Risk*. Germany, Springer Berlin Heidelberg, 2015.
6. UNISDR. www.unisdr.org (Online access to various UNISDR documents and reports)
7. Wisner B, Blaikie P., Cannon T. and I. Davis. *At Risk: Natural Hazards, People's Vulnerability, and Disasters,* 2nd Edition, Routledge, London, 2014.

Suggested Readings

1. Bird, R. (ed.). *Handbook of Natural Hazards and Disasters*. United States, Larsen and Keller Education, 2022.
2. Keller, Edward A and Duane E DeVecchio. *Natural Hazards: Earth's Processes as Hazards, Disasters, and Catastrophes*. 4th Edition, Routledge, 2016.
3. Madu, Christian N and Chu-hua Kuei. *Handbook of Disaster Risk Reduction and Management*. World Scientific Publishing Co. Pvt. Ltd, Singapore, 2018.
4. Montz, Burrell E.; Graham A. Tobin and Ronald R. Hagelman. *Natural Hazards: Explanation and Integration*. Second Edition, The Guilford Press, New York, 2017.
5. Pittock, A. Barrie. *Climate Change: The Science, Impacts and Solutions*. United Kingdom, Taylor & Francis, 2009.

DM 106
MAPPING DISASTERS IN INDIA

Course Code: DM 106	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical/Lab Work	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Identifyand describe disasters prone areas in India.
- Compare and explain spatial variations in disaster manifestation.
- Analyse, demarcate and corelate disaster risk zones.
- Measure and compute disaster risk potential and vulnerability.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Preparing Report on Disaster Patterns and Trend in India
2. Mapping Earthquake Prone Regions in India
3. Analysing India's Flood Prone Regions
4. Cyclone Prone Zones and Affected Areas
5. Landslide Hazard Zonationof India
6. Demarcating Drought Prone Regions of India
7. Mapping Multi-Hazard Prone Districts of India

LIST OF READINGS

Essential Readings:

1. Gupta, M.C.; L.C. Gupta, B.K. Tamini, Vinod K. Sharma: *Manual on Natural Disaster Management in India,* National *Disaster Management Centre, New Delhi, 2000.*
2. Kapur, Anu. Vulnerable India: A Geographical Study of Disasters. India, SAGE, 2010.
3. Peijun Shi, Roger E. Kasperson (eds.). *World Atlas of Natural Disaster Risk*. Germany, Springer Berlin Heidelberg, 2015.
4. Valdiya, K.S. *Coping with Natural Hazards: Indian Context*. India, Orient Longman, 2004.
5. Subramanian, R.: *Disaster Management*, Vikas Publishing House Pvt. Ltd., Noida, India, 2018.

Suggested Readings

1. Alexander, D. E.: *Natural Disasters*. London: University College London Press and; Dordrecht and Boston: Kluwer Academic Publishers, 1993.
2. Alexander, D. E.: *Confronting Catastrophe: New Perspectives on Natural Disasters*. Harpenden, U.K: Terra Publishing, 2000.
3. Hewitt, K. *Regions of Risk: A Geographical Introduction to Disasters*. United Kingdom, Taylor & Francis, 2014.
4. Wisner B, Blaikie P., Cannon T. and I. Davis. *At Risk: Natural Hazards, People's Vulnerability, and Disasters,* 2nd Edition, Routledge, London, 2014.

DM 107
DISASTER EMERGENCY MANAGEMENT SKILLS

Course Code: DM 107	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical/Lab Work	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Learn disaster emergency management skills
- Conduct and execute TRIAGE activities.
- Apply First Aid and CPR during emergency situation.
- Deliver emergency and rescue services

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSECONTENT: List of Exercises

1. Hands on Training on TRIAGE
2. First Aid: Wound, Bleeding, Burn, Management
3. Securing Physical Injuries: Splinting Techniques
4. Heimlich Maneuver Technique
5. Performing Cardiopulmonary Resuscitation (CPR) Techniques

LIST OF READINGS

Essential Readings:

1. Abarquez I., Murshed Z. *Field Practitioners' Handbook- Community-based Disaster Risk Management*, Asian Disaster Preparedness Centre, Bangkok, Thailand, 2004.
2. Kelman, I., Mercer, J. and Gillard, J C. *The Routledge Handbook of Hazards and Disaster Risk Reduction Including Climate Change Adaptation*, Routledge, 2017.
3. Madu, Christian N and Chu-hua Kuei. *Handbook of Disaster Risk Reduction and Management*. World Scientific Publishing Co. Pvt. Ltd, Singapore, 2018.
4. National Institute of Disaster Management. *India Disaster Report 2011*. New Delhi, 2012.
5. United Nations Development Programme. *Disaster Management in India, 2004*.
6. http://nidm.gov.in/ (Online access to guidelines and manuals prepared by NIDM)

Suggested Readings:

1. Abarquez I., Murshed Z. *Field Practitioners' Handbook- Community-based Disaster Risk Management*, Asian Disaster Preparedness Centre, Bangkok, Thailand, 2004.
2. Coppola D.P. *Introduction to International Disaster Management*, 2015, Elsevier, UK.
3. Kelman, I., Mercer, J. and Gillard, J C. *The Routledge Handbook of Hazards and Disaster Risk Reduction Including Climate Change Adaptation*, Routledge, 2017.
4. Shaw, R (ed). *Community- Based Disaster Risk Reduction*, Emerald Group Pub. Limited, UK, 2012.
5. Wisner, B, JC Gaillard and I. Kelman. *The Routledge Handbook of Hazards and Disaster Risk Reduction*, Routledge, Taylor and Francis Group, Oxon, 2012.

DM 108
STATISTICAL ANALYSIS

Course Code: DM 108	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical/Lab Work	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of this course, the students will be able to:

- Comprehend statistical techniques and their applications.
- Analyse and synthesize various types of spatial and non-spatial datasets.
- Measure and apply descriptive and inferential statistical techniques.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examination.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: List of Practical/Lab Exercises:

1. Data: Spatial & Non-spatial, Grouped &Ungrouped, Continuous &Discrete
2. Descriptive Statistics
    a. Measures of Central Tendency: Mean, Median and Mode
    b. Measures of Dispersion: Range, variance, standard deviation, coefficient of variability
3. Normal Distribution: Characteristics, measurements and applications; z-distribution

4. Inferential Statistics:
    a. Parametric statistical test: T-Test, Z-Testand ANOVA
    b. Non-parametric statistical test: Chi-Square Test
    c. Correlation: Scatterplot, Spearman's method and Pearson's correlation
    d. Regression analysis, construction of regression line.

LIST OF READINGS

Essential Readings:

1. Flora, David B. *Statistical Methods for the Social and Behavioural Sciences: A Model-Based Approach*. United Kingdom, SAGE Publications, 2017.
2. Gupta, S. P.: *Statistical Methods*, Sultan Chand and Sons, 2014.
3. Ott, R. Lyman and Michael Longnecker: *An Introduction to Statistical Methods and Data Analysis*, Brooks/Cole, USA, 2010.
4. Prasad, Sahana. *Elementary Statistical Methods*. Germany, Springer Nature Singapore, 2023.
5. Willard, Cheryl Ann. *Statistical Methods: An Introduction to Basic Statistical Concepts and Analysis*. United Kingdom, Taylor & Francis, 2020.

Suggested Readings:

1. Burt, J.E., G.M. Barber and D.L. Rigby: *Elementary Statistics for Geographers*, Guilford Press, New York, 2009.
2. Hanneman, Robert A., et al. *Basic Statistics for Social Research*. Germany, Wiley, 2012.
3. Rogerson, Peter. A.: *Statistical Methods for Geography: A Student's Guide*, Sage Publications, London, 2015.
4. Sarkar, Ashis: *Quantitative Geography: Techniques and Presentations*, Orient Blackswan, United States, 2013.
5. Taylor, P.J.: *Quantitative Methods in Geography*,Houghton Mifflin Company, Boston, 1977.

SEMESTER II

DM 201
DISASTER MANAGEMENT MECHANISMS IN INDIA

Course Code: DM 201	**Max. Marks: 100**
Course Type: DSC- Theory	Theory Paper: 80
Course Credit: 4	Internal Assessment: 20
Teaching Hours/Week: 4	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Comprehended and appreciate India's approach to disaster risk reduction.
- Outline institutional, financial and legal framework for disaster management.
- Evaluate and Critique India's disaster risk managementpolicy and plans.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each part shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to the compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit I

i. Disaster-scape of India: An Overview
ii. Role of Planning Commission in Disaster Management, DM Act, India

Unit II

iii. Disaster Management Policy in India
iv. Institutional, Financial and Legal Frameworks for Disaster Management

Unit III

v. Role and Responsibilities of NDMA, SDMA, DDMA
vi. Role of NIDM and SIDM; Disaster Response Forces: Role of NDRF, SDRF

Unit IV

vii. Role of Municipalities/Urban Local Bodies in Disaster Management
viii. Contribution and functions of Panchayati Raj Institutions

LIST OF READINGS

Essential Readings:

1. Government of India. *Disaster Management Act (2005)*, Ministry of Home Affairs, 2005.
2. Goel, S.L. *Encyclopaedia of Disaster Management*, Deep & Deep Pub., New Delhi, 2006.
3. Gosh, G.K. *Disaster Management*, A.P.H. Publishing Corporation, New Delhi, 2016.
4. Government of India. *Disaster Management in India - A Status Report*, 2004.
5. Government of India. *Disaster Management in India*, 2005.

Suggested Readings:

1. Chaturvedi, S., Guleria, S. and Bindal, M.K. *India Disaster Report (2014-17).* National Institute of Disaster Management (NIDM), New Delhi. 2020.
2. National Disaster Management Authority. *National Policy on Disaster Management, 2009*.
3. National Institute of Disaster Management. *India Disaster Report 2011*. New Delhi, 2012.

4. Shastri, K.N. *Disaster Management in India*, 2010, Pinnacle Technology.
5. United Nations Development Programme. *Disaster Management in India, 2004*.
6. http://nidm.gov.in/ (Online access to guidelines and manuals prepared by NIDM)

DM 202
GLOBAL INITIATIVES FOR DRR AND CLIMATE RESILIENCE

Course Code: DM 202	**Max. Marks: 10**
Course Type: DSC- Theory	Theory Paper: 80
Course Credit: 4	Internal Assessment: 20
Teaching Hours/Week: 4	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Recognise, identify and review international initiatives for DRR and Climate resilience
- Analyse and compare the outcomes of global efforts for risk reduction and climate change
- Develop ideas to formulate appropriate action framework for risk management
- Critique the actions taken and success of global efforts for DRR and climate resilience

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each part shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to the compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

Global Initiatives for Disaster Risk Reduction

i. Background to Global Initiatives for Disaster Management
ii. IDNDR and ISDR: Formulation, Priorities and Outcomes
iii. Hyogo Framework (HFA): Framework, Priorities for Action and outcomes

Unit-II

Contemporary Disaster Risk Reduction Framework and Future Agenda

iv. Sendai Framework: Goals, Guiding Principles and Priorities for Action
v. UNDRR- SEM and UNDRR Partnership and Stakeholder Strategy,
vi. International Agencies for Disaster Management

Unit-III

Global Initiatives for Climate Change

vii. Overview: Climate Initiatives since 1970s
viii. Rio-Earth Summit, Montreal and Kyoto Protocol
ix. IPCC- Formulation, vision, activities and outcomes

Unit-IV

International Frameworks for Climate Change Resilience

x. UNFCCC- Objectives and achievements; COP
xi. Paris Agreement; Action for Climate Empowerment (ACE)
xii. India's Initiatives: NAPCC, OSOWOG; Climate Change Policy

LIST OF READINGS

Essential Readings:

1. Coppola, D. P. *Introduction to International Disaster Management*. 3rd Ed. Elsevier, London, 2015.
2. Elliott, A. and E.V. Hsu (eds.). *The Consequences of Global Disasters*, Routledge, London, 2016.
3. Hardy, John T. *Climate Change: Causes, Effects, and Solutions*. United Kingdom, Wiley, 2003.
4. IFRC. https://www.ifrc.org/documents/all (Online access to various World Disaster Reports)
5. IPCC. https://www.ipcc.ch/reports/ (For reports and information on climate change)
6. The United Nations Office for Disaster Risk Reduction (UNISDR). *Sendai Framework for Disaster risk Reduction: 2015-2030*, Geneva, Switzerland, 2015.
7. Tim Cadman, Rowena Maguire, and Charles Sampford (eds.). *Governing the Climate Change Regime: Institutional Integrity and Integrity Systems*. United Kingdom, Taylor & Francis, 2016.
8. UNFCC. https://unfccc.int/ (For documents, reports and information on climate change actions)
9. UNISDR. www.unisdr.org (Online access to various UNISDR documents and reports)
10. Vogler, John. *Climate Change in World Politics*. United Kingdom, Palgrave Macmillan UK, 2016.

Suggested Readings:

1. Daniel P. Shepardson, Anita Roychoudhury, Andrew S. Hirsch. *Teaching and Learning about Climate Change: A Framework for Educators*. United Kingdom, Taylor & Francis, 2017.
2. Khan, Mizan R, et al. *The Paris Framework for Climate Change Capacity Building*. United Kingdom, Taylor & Francis, 2018.
3. Madu, Christian N and Chu-hua Kuei. *Handbook of Disaster Risk Reduction and Management*. World Scientific Publishing Co. Pvt. Ltd, Singapore, 2018.
4. Montz, Burrell E.; Graham A. Tobin and Ronald R. Hagelman. *Natural Hazards: Explanation and Integration*. Second Edition, The Guilford Press, New York, 2017.
5. Vesselin Popovski (Ed.). *The Implementation of the Paris Agreement on Climate Change*. United Kingdom, Taylor & Francis, 2018.
6. Wisner B, Blaikie P., Cannon T. and I. Davis. *At Risk: Natural Hazards, People's Vulnerability, and Disasters,* 2nd Edition, Routledge, London, 2014.

DM 203
FUNDAMENTALS OF GEOINFORMATICS

Course Code: DM 203	**Max. Marks: 50**
Course Type: DSC- Theory	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Comprehend nature and concepts of Geoinformatics and its application.
- Analyse geoinformatics based data sources and extract information.
- Apply and execute tool and techniques of remote sensing, GIS and GPS systems.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to the compulsory question at S. No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit I

Fundamental Concepts of Remote Sensing

i. Remote Sensing: Concept, Types and Applications
ii. Platforms, Sensors and Scanning Systems; Electromagnetic Spectrum

Unit-II

Satellite Image Interpretation

iii. Satellite Image: Meaning and Types
iv. Resolution: Spatial, Spectral, Radiometric and Temporal; Elements of Image Interpretation.

Unit-III

Fundamentals of Geographic Information Systems:

v. GIS: Concept, Significance, Component and Applications
vi. Spatial data base: Types and Representations; GIS Operations: Buffer, Overlay and Query

Unit IV

Fundamentals of Global Navigational Satellite System

i. GNSS Concept, Principles of Operations, GPS Segments
ii. GPS Positioning, GPS Accuracy and Errors; Applications of GNSS

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne. *Introduction to Remote Sensing* (5thEd.), Guilford Press, 2012.
2. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
3. Gomarasca, Mario A. *Basics of Geomatics*, Springer: Heidelberg, 2009.
4. Kang-Tsung Chang.*Introduction to Geographic Information Systems*, Tata McGraw-Hill, New Delhi, 2015.
5. Srivastava, G.S. *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014

Suggested Readings:

1. Burrough, P. A., McDonnell, R. A. and Lloyd, C.D. *Principles of Geographical Information Systems*, 3rd. ed., Oxford University Press, Toronto, 2016.
2. Wise, S. *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.
3. Leick, A, Lev Rapoport and Dmitry Tatarnikov.*GPS Satellite Surveying*, John Wiley & Sons, 2015.
4. Longley, Paul A., Goodchild, Michael F. Maguire, David J., and David W. Rhind.*Geographic Information Systems and Science*, 3rd edition., John Wiley and Sons, England, 2011.
5. Heywood, I. Conrnelius, S. and Carver, S.*An Introduction to Geographical Information Systems*, Pearson Education Limited, United Kingdom. 2010.

DM 204
RESEARCHMETHODOLOGYINDISASTER STUDIES

Course Code: DM 204	***Evaluation: Maximum Marks: 50***
Course Type: RBC-RM	Terminal Examination: 40 Marks
Course Credit: 2	Internal Assessment: 10 Marks
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the semester, students will be able to:

- Understand the principles of research and research process.
- Develop knowledge on methods and techniques in conducting research.
- Design a research proposal to address a specified problem.

- ❖ Establish and follow research ethics while executing research.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 6 parts in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-1

Basic Concepts in Research:

i. Meaning and objectives of research
ii. Research types and approaches; Geographic perspective in research

Unit-II

Initiating and Conceptualising Research:

iii. Research process; Conceptual and Theoretical Framework
iv. Research problem: Selection and techniques; Research design: need and features

Unit-III

Data Collection Tools and Methods:

v. Types and sources of data; Primary data collection- Fieldwork, observation and survey
vi. Consideration of ethical issues in data collection

Unit-IV

Data Analysis and Report writing

vii. Use of statistics in research
viii. Elements of report writingincluding internship report; Referencing and bibliography

LIST OF READINGS

Essential Readings:

1. Creswell, John W., *Research Design; Qualitative, Quantitative and Mixed Methods Approach*, SAGE Publications, Los Angeles, 2008.
2. Kothari, C. R. *Research Methodology: Methods and Techniques*. India, New Age International Pvt Limited Publishers, 2013.
3. Stallings, Robert A. *Methods of Disaster Research*. United States, Xlibris US, 2003.
4. Jason D. Rivera. *Disaster and Emergency Management Methods: Social Science Approaches in Application.* United Kingdom, Taylor & Francis, 2021.
5. Ozerdem, A. and Bowd, R. (eds). *Participatory Research Methodologies: Development and Post-Disaster/Conflict Reconstruction*. United Kingdom, Taylor & Francis, 2016.
6. Townes, J.A. and Decker, E.N. *Handbook of Research on Disaster Management and Contingency Planning in Modern Libraries*. United States, Information Science Reference, 2016.
7. Ricci, Edmund M., et al. *Disaster Evaluation Research: A Field Guide*. United Kingdom, Oxford University Press, 2019.

Suggested Readings:

1. Flick, U. *An Introduction to Qualitative Research*, 5th Edition, SAGE, 2014.

2. Kumar Ranjit, *Research Methodology: A step-by-step Guide for Beginners*, SAGE Publications, Ltd. London (Third Edition), 2010.
3. Montello, D. and P. Sutton, *An Introduction to Scientific Research Methods in Geography and Environmental Studies*, SAGE, 2012
4. Misra, H.N. and V.P. Singh, *Research Methodology in Geography*, Rawat Pub., Jaipur, 1998.
5. Misra, R.P., *Research Methodology*, Concept Publishing Company, New Delhi, 1989.

DM 205
RESEARCH TOOLS AND TECHNIQUES

Course Code: DM 205	***Evaluation: Maximum Marks: 50***
Course Type: RBC-RM (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- ❖ Generate primary data for research using various tools and techniques.
- ❖ Construct measurement tools for research problems.
- ❖ Develop ability for authentic, coherent and lucid writing styles.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSECONTENT: List of Exercises

1. Writing literature review and identifying research gaps
2. Research design on selected theme
3. Sampling and spatial sampling techniques for data collection
4. Preparation of questionnaire
5. Preparation of interview schedule
6. Using measurement tools: Scaling techniques and cumulative index
7. Reference Stylesand bibliography
8. Performing similarity checks and plagiarism

LIST OF READINGS

Essential Readings:

1. Creswell, John W., *Research Design; Qualitative, Quantitative and Mixed Methods Approach*, SAGE Publications, Los Angeles, 2008.
2. Kothari, C. R. *Research Methodology: Methods and Techniques*. India, New Age International Pvt Limited Publishers, 2013.
3. Stallings, Robert A. *Methods of Disaster Research.* United States, Xlibris US, 2003.
4. Jason D. Rivera. *Disaster and Emergency Management Methods: Social Science Approaches in Application*. United Kingdom, Taylor & Francis, 2021.
5. Ricci, Edmund M., et al. *Disaster Evaluation Research: A Field Guide.* United Kingdom, Oxford University Press, 2019.

Suggested Readings:

1. Flick, U. *An Introduction to Qualitative Research*, 5th Edition, SAGE, 2014.
2. Kumar Ranjit, *Research Methodology: A step-by-step Guide for Beginners*, SAGE Publications, Ltd. London (Third Edition), 2010.
3. Montello, D. and P. Sutton, *An Introduction to Scientific Research Methods in Geography and Environmental Studies*, SAGE, 2012
4. Ozerdem, A. and Bowd, R. (eds). *Participatory Research Methodologies: Development and Post-Disaster/Conflict Reconstruction.* United Kingdom, Taylor & Francis, 2016.

DM 206
DISASTER PREPAREDNESS AND CAPACITY BUILDING

Course Code: DM 206	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks

COURSE OUTCOMES: At the end of the course, the students will be able to:

- ❖ Learn disaster preparednesspractices and protocols.
- ❖ Prepare tabletop and functional exercises for capacity building.
- ❖ Develop skills to conduct and execute mock drills.
- ❖ Accomplish capacity building through emergency management and planning.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSECONTENT: List of Exercises

1. Mock Drill for Earthquake Safety and Preparedness
2. Fire Safety Drills and Planning
3. Learning Usages and application of Fire Extinguishers
4. School Safety Planning and Management
5. Disaster Emergency and Evacuation Planning
6. Crowd Management and Planning

LIST OF READINGS

Essential Readings:

1. Abarquez I., Murshed Z. *Field Practitioners' Handbook- Community-based Disaster Risk Management*, Asian Disaster Preparedness Centre, Bangkok, Thailand, 2004.
2. Kelman, I., Mercer, J. and Gillard, J C. *The Routledge Handbook of Hazards and Disaster Risk Reduction Including Climate Change Adaptation*, Routledge, 2017.
3. Madu, Christian N and Chu-hua Kuei. *Handbook of Disaster Risk Reduction and Management*. World Scientific Publishing Co. Pvt. Ltd, Singapore, 2018.
4. National Institute of Disaster Management. *India Disaster Report 2011*. New Delhi, 2012.
5. United Nations Development Programme. *Disaster Management in India, 2004*.
6. http://nidm.gov.in/ (Online access to guidelines and manuals prepared by NIDM)

Suggested Readings:

1. Abarquez I., Murshed Z. *Field Practitioners' Handbook- Community-based Disaster Risk Management*, Asian Disaster Preparedness Centre, Bangkok, Thailand, 2004.
2. Coppola D.P. *Introduction to International Disaster Management*, 2015, Elsevier, UK.
3. Kelman, I., Mercer, J. and Gillard, J C. *The Routledge Handbook of Hazards and Disaster Risk Reduction Including Climate Change Adaptation*, Routledge, 2017.
4. Shaw, R (ed). *Community- Based Disaster Risk Reduction*, Emerald Group Pub. Limited, UK, 2012.
5. Wisner, B, JC Gaillard and I. Kelman. *The Routledge Handbook of Hazards and Disaster Risk Reduction*, Routledge, Taylor and Francis Group, Oxon, 2012.

DM 207
GEOSPATIAL TOOLS AND TECHNIQUES

Course Code: DM 207	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Apply and handle geospatial tools for data collection, analysis and interpretation.
- Collect and create accurate and authenticate geodatabase using geospatial platforms.
- Execute spatial analysis tools and techniques to produce meaningful data output.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examination.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: ***List of Themes/Exercises***

1. Imagery Processing: Georeferencing
2. Preparing Mosaic and Subset
3. Interpretation of Features from Digital Satellite Imageries
4. GIS Software: Georeferencing
5. Digitization and Geodatabase Creation
6. GIS Operations: Buffer, Query and Overlay Operation
7. Unsupervised and Supervised Classification
8. Map Designing: Creating Map Layouts
9. GNSS: Data Collection and Survey
10. GPS Data Processing and Mapping

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne. *Introduction to Remote Sensing* (5thEd.), Guilford Press, 2012.
2. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
3. Gomarasca, Mario A. *Basics of Geomatics*, Springer: Heidelberg, 2009.

4. Kang-Tsung Chang.*Introduction to Geographic Information Systems*, Tata McGraw-Hill, New Delhi, 2015.
5. Longley, Paul A., Goodchild, Michael F. Maguire, David J., and David W. Rhind.*Geographic Information Systems and Science*, 3rd edition., John Wiley and Sons, England, 2011.

Suggested Readings:

1. Brewer, C.A. *Designing Better Maps: A Guide for GIS Users*, ESRI Press, 2005.
2. DeMeers, M.N.*Fundamentals of Geographic Information Systems*, John Wiley and Sons, 2008.
3. Dong, P. and Qi Chen.*LiDAR Remote Sensing and Applications*. Taylor & Francis Group, 2018
4. Srivastava, G.S. *An Introduction to Geoinformatics*, McGraw Hill Education, New Delhi, 2014
5. Wise, S. *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.

DM 208
FIELD-WORK BASED ACTIVITY

Course Code: DM 208	***Evaluation: Maximum Marks: 50***
Course Type: DSC- Practical	Field-Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- ❖ Recognize the importance of field-based activities in disaster management
- ❖ Learn skills for social-work in disaster risk assessment
- ❖ Measure and analyse people's perception of hazards and disaster
- ❖ Perform field-work based auditing of DRR plans

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Field-work based record under the supervision and guidance of class teacher. The candidates shall submit activity record file duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSECONTENT: List of Exercises

1. Learning from the Experts-I: Social Work in Disaster Management
2. Learning from the Experts-II: Psychosocial First Aid (PSFA) in disaster management
3. Field Data Collection: Hazard and Disaster Management Perception
4. Undertaking Disaster Preparedness Auditing
5. Preparing Institutional Disaster Management and Risk Reduction Plan

LIST OF READINGS

Essential Readings:

1. Creswell, John W., *Research Design; Qualitative, Quantitative and Mixed Methods Approach*, SAGE Publications, Los Angeles, 2008.
2. Kothari, C. R. *Research Methodology: Methods and Techniques*. India, New Age International Pvt Limited Publishers, 2013.
3. Stallings, Robert A. *Methods of Disaster Research.* United States, Xlibris US, 2003.

4. Jason D. Rivera. *Disaster and Emergency Management Methods: Social Science Approaches in Application*. United Kingdom, Taylor & Francis, 2021.
5. Ricci, Edmund M., et al. *Disaster Evaluation Research: A Field Guide.* United Kingdom, Oxford University Press, 2019.

Suggested Readings:

1. Flick, U. *An Introduction to Qualitative Research*, 5th Edition, SAGE, 2014.
2. Kumar Ranjit, *Research Methodology: A step-by-step Guide for Beginners*, SAGE Publications, Ltd. London (Third Edition), 2010.
3. Montello, D. and P. Sutton, *An Introduction to Scientific Research Methods in Geography and Environmental Studies*, SAGE, 2012
4. Ozerdem, A. and Bowd, R. (eds). *Participatory Research Methodologies: Development and Post-Disaster/Conflict Reconstruction.* United Kingdom, Taylor & Francis, 2016.